

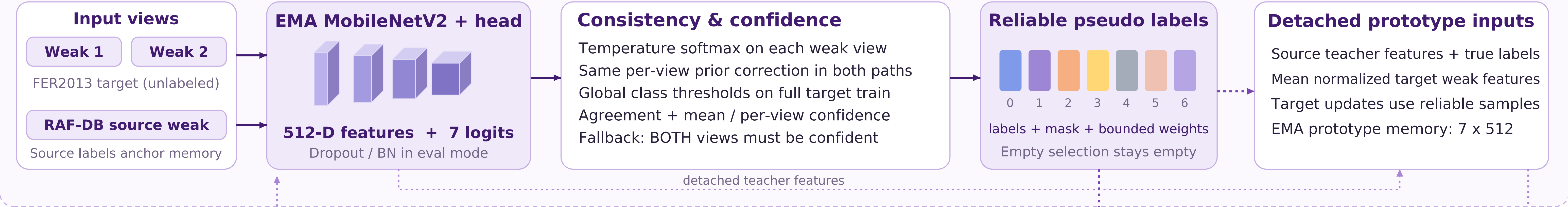
MobileNetV2 + Dual-View EMA CAST

CODE-AUDITED FRAMEWORK

RAF-DB to FER2013 | Target adaptation after source pre-training | Re-read from GitHub commit c67ac74

2. Dual-View EMA Teacher

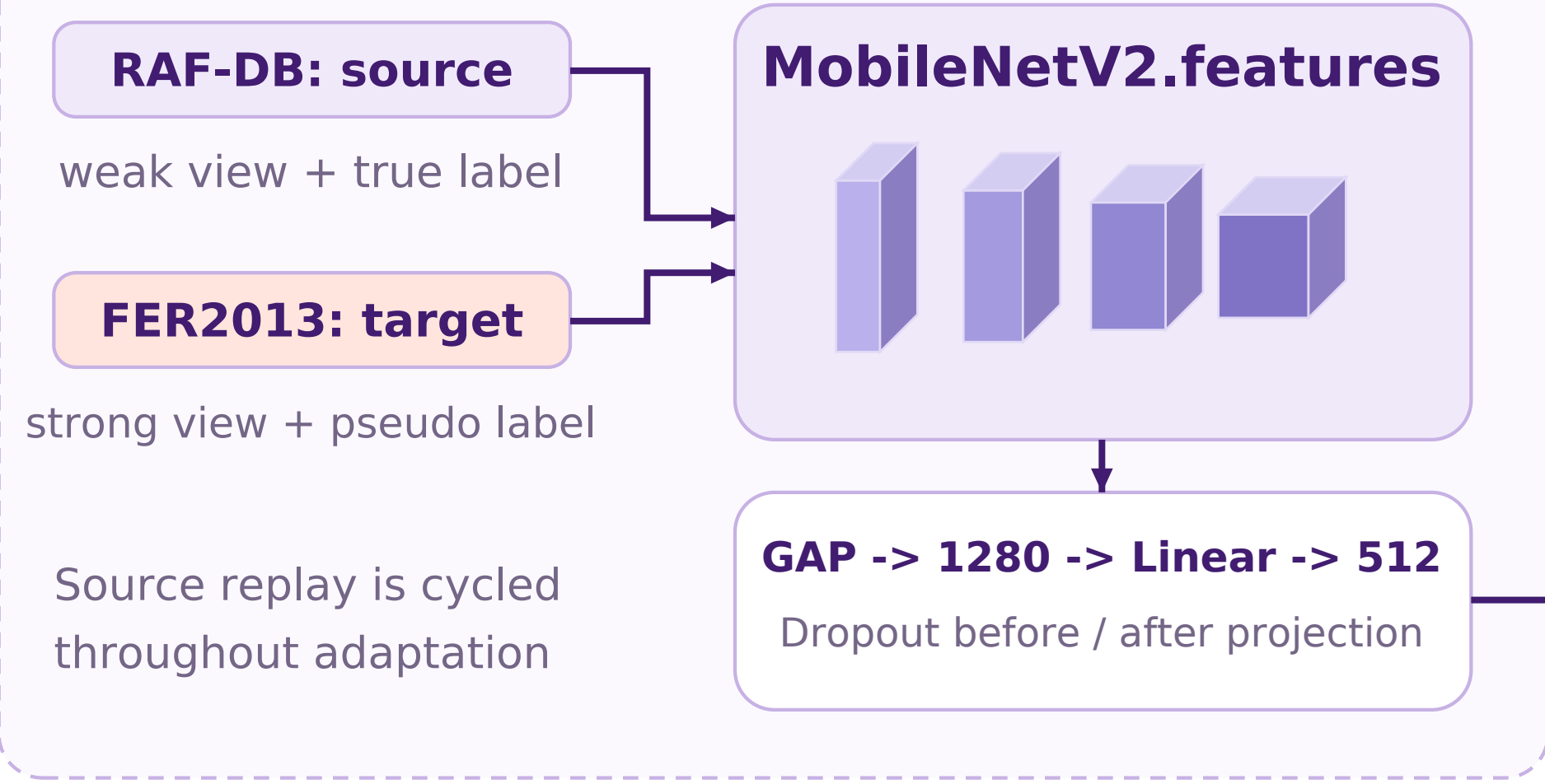
Independent copy of the entire student | eval mode | no gradient



Student -> teacher EMA
after each optimizer step

1. Shared Student Backbone

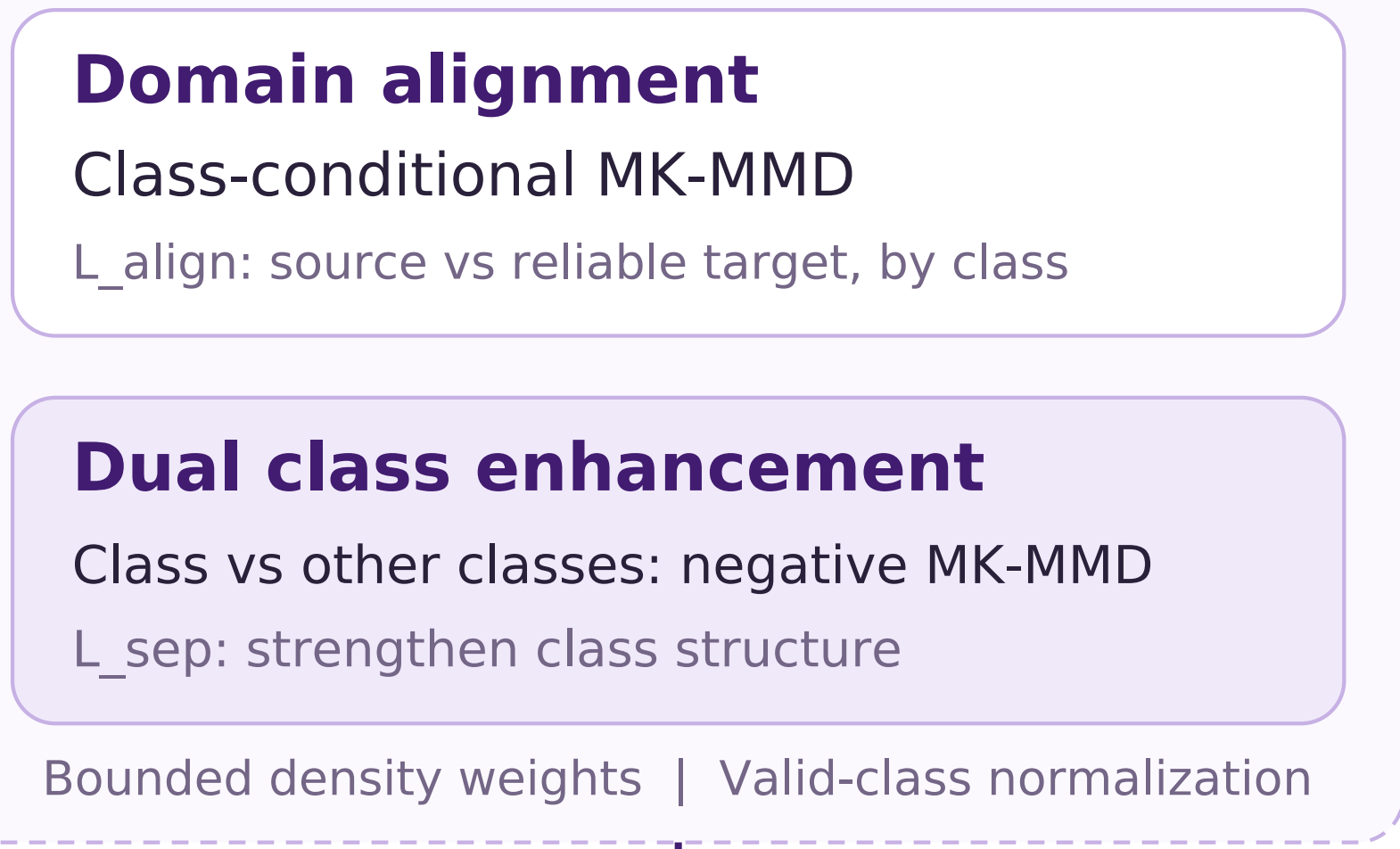
Source weak view + target strong view share parameters



Backpropagate to student

3. DDRL Module

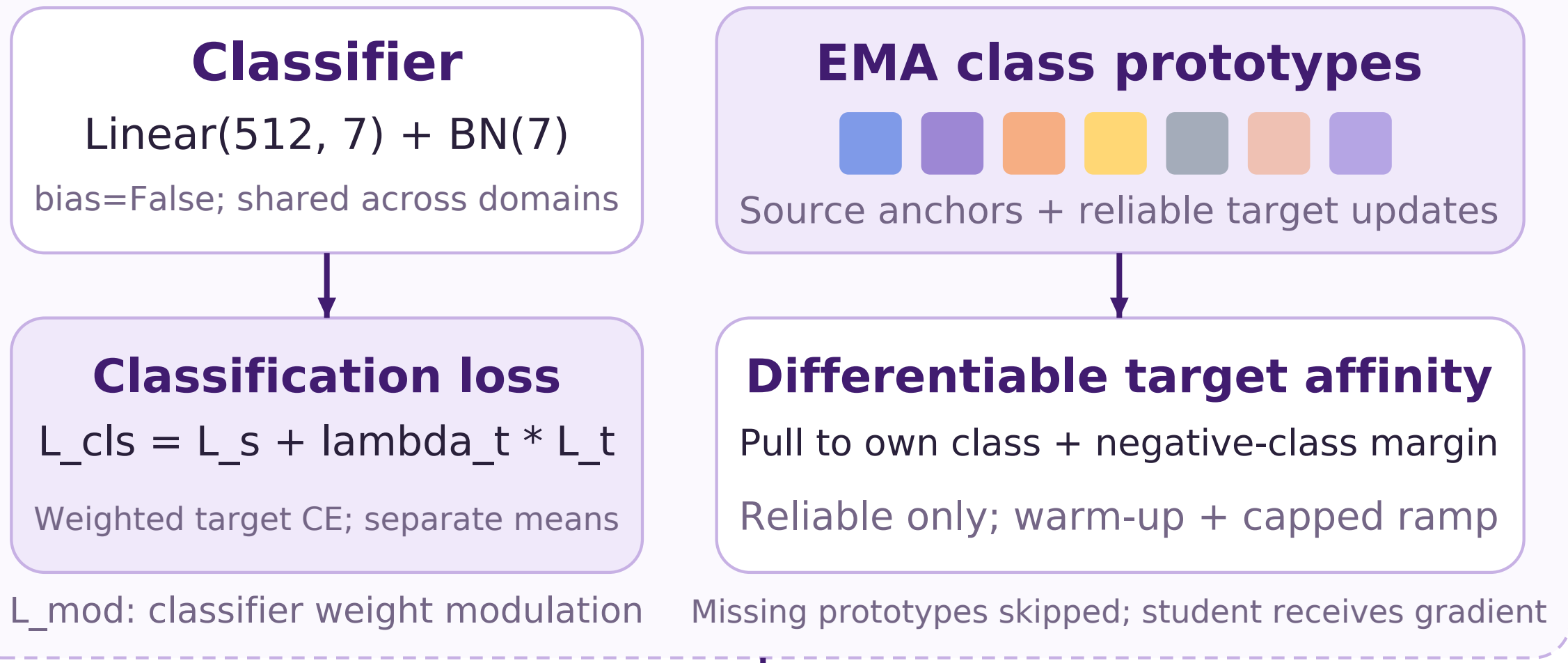
Loss constraints on the same 512-D student features



DDRL loss

4. CCDR Module

Classifier regularization + stable target affinity



Classification + affinity + modulation

TOTAL TARGET-STAGE TRAINING OBJECTIVE

$$L = w1 (L_s + \lambda_t L_t) + w2 (L_{align} + L_{sep}) + \lambda_{aff}(\text{epoch}) L_{aff} + w3 L_{mod}$$

→ Forward / loss input > Pseudo-label supervision > EMA / memory (no gradient) - - -> Gradient update Inference: student backbone + classifier only